Mirpur University of Science & Technology

Department of Software Engineering

SOFTWARE QUALITY ENGINEERING

Lab Assignment #2

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FA23-BSE-039

SECTION: A

Report:

GitHub repo link: https://github.com/Maryam-Ikhlaq-24/SQE-Labs-Maryam-Ikhlaq-24

SECTION 1 — Cover Information

App 1: Google Chrome
Version: 147.0.7727.111
Platform: Android
Device: Samsung Galaxy A16
Evaluator: Maryam Ikhlaq
Student ID: FA23-BSE-039

App 2: Microsoft Edge
Version: 147.0.3912.77
Platform: Android
Device: Samsung Galaxy A16
Evaluator: Maryam Ikhlaq
Student ID: FA23-BSE-039

SECTION 2 — App Overviews

Google Chrome is a widely used mobile web browser developed by Google, designed to provide fast, secure, and seamless internet browsing. It targets a broad audience, including general users, students, and professionals. Key features include tab management, synchronization across devices using a Google account, password management, and integration with Google services. Chrome emphasizes speed, reliability, and simplicity.

Microsoft Edge is a modern web browser developed by Microsoft, also targeting a wide user base. It provides standard browsing capabilities along with additional features such as tracking prevention, built-in ad blocking, and read-aloud functionality. Edge focuses more on privacy and productivity features while maintaining competitive performance. Both applications serve the same primary purpose but differ slightly in design philosophy, with Chrome focusing on simplicity and Edge offering more built-in tools.

SECTION 3 — Evaluation Methodology

The evaluation was conducted using the ISO/IEC 25010 software quality model, which includes eight characteristics: Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability. Each characteristic was rated on a Likert scale from 1 (Unacceptable) to 5 (Excellent).

Both applications were tested on the same device (Samsung A16, Android 15) under similar conditions to ensure fairness. Testing involved real-world usage scenarios such as browsing websites, switching tabs, sharing content, and operating under limited network conditions. Performance metrics were measured using a stopwatch, while battery and data usage were observed through device settings. Reliability was evaluated through continuous usage and offline testing, and security was assessed by reviewing permissions, encryption, and authentication mechanisms.

SECTION 4 — Completed Checklists (Summary)

App 1 — Google Chrome

FULL EVALUATION CHECKLIST				
Quality Characteristic (ISO 25010)	**Rating (1–5)**	**Weight**	***Observations / Evidence (min. 2 specific points)***	**Weighted Score**
1. Functional Suitability	**5**	15%	*Successfully loaded multiple websites (Google, YouTube, Wikipedia) without rendering errors.* *Tab management and bookmark sync worked correctly across sessions using Google account.*	**0.75**
2. Performance Efficiency	**4**	12%	*Average page load time measured ~0.8–1.2 seconds on WiFi using stopwatch.* *Battery usage increased by ~3–4% after 10 minutes of continuous browsing.*	**0.48**
3. Compatibility	**5**	10%	*Data (history, bookmarks) synced instantly between mobile and desktop Chrome.* *Sharing links from Chrome to apps like WhatsApp and Gmail worked without failure.*	**0.50**
4. Usability	**4**	15%	*Primary tasks (open site, switch tab, share link) completed within 1–5 seconds without errors.* *Settings menu required multiple steps to locate advanced options (slight usability friction).*	**0.60**
5. Reliability	**5**	15%	*No crashes or freezes observed during continuous 10-minute browsing session.* *Tabs remained intact and restored correctly after force-closing the application.*	**0.75**
6. Security	**5**	15%	*Displayed "Secure (HTTPS)" indicator for encrypted websites.* *Google Safe Browsing warned against potentially harmful sites during testing.*	**0.75**
7. Maintainability	**5**	8%	*App received frequent updates on Play Store (bi-weekly updates observed).* *Release notes clearly mentioned bug fixes and performance improvements.*	**0.40**
8. Portability	**5**	10%	*Chrome installed and ran successfully on multiple platforms (Android, Windows).* *User data (bookmarks, passwords) synchronized automatically across devices.*	**0.50**
WEIGHTED TOTAL SCORE (out of 5.00)				**4.73**

Functional Suitability: 5 (All features working correctly, complete browsing functionality)

Performance Efficiency: 4 (Fast performance, slight battery consumption)

Compatibility: 5 (Supports all major platforms and syncs effectively)

Usability: 4 (Easy to use, minor complexity in settings navigation)

Reliability: 5 (Stable with no crashes and strong recovery)

Security: 5 (HTTPS, safe browsing, secure authentication)

Maintainability: 5 (Frequent updates and fast bug fixes)

Portability: 5 (Excellent cross-device synchronization)

Weighted Score: 4.7 / 5.00

App 2 — Microsoft Edge

FULL EVALUATION CHECKLIST				
Quality Characteristic (ISO 25010)	**Rating (1–5)**	**Weight**	**Observations / Evidence (min. 2 specific points)**	**Weighted Score**
1. Functional Suitability	**5**	15%	*Loaded all tested websites correctly with no broken layouts or missing elements.* *Built-in features like read-aloud and ad-blocking functioned as expected.*	**0.75**
2. Performance Efficiency	**4**	12%	*Page load times measured ~0.9–1.3 seconds under similar conditions.* *Battery consumption observed at ~3% during 10-minute usage session.*	**0.48**
3. Compatibility	**5**	10%	*Sync worked correctly across devices using Microsoft account (bookmarks and tabs).* *Successfully shared web pages to external apps like WhatsApp and Messenger.*	**0.50**
4. Usability	**4**	15%	*Basic tasks (search, open tabs, share links) completed efficiently within seconds.* *Interface slightly cluttered due to multiple built-in tools, affecting navigation clarity.*	**0.60**
5. Reliability	**5**	15%	*No crashes occurred during testing period.* *App handled Airplane mode properly by showing offline message and recovered after reconnect.*	**0.75**

6. Security	**5**	15%	*Tracking prevention feature actively blocked trackers during browsing.* *HTTPS encryption used consistently for secure connections.*	**0.75**
7. Maintainability	**4**	8%	*Updates available regularly but less frequent than Chrome (monthly pattern observed).* *Changelog includes general fixes but less detailed compared to Chrome.*	**0.32**
8. Portability	**5**	10%	*Available across Android, Windows, and other platforms with seamless installation.* *Sync feature maintained consistency of bookmarks and browsing data across devices.*	**0.50**
WEIGHTED TOTAL SCORE (out of 5.00)				**4.65**

Functional Suitability: 5 (Complete functionality with additional features)

Performance Efficiency: 4 (Smooth performance comparable to Chrome)

Compatibility: 5 (Strong cross-platform support)

Usability: 4 (Good UI but slightly cluttered)

Reliability: 5 (Stable and consistent performance)

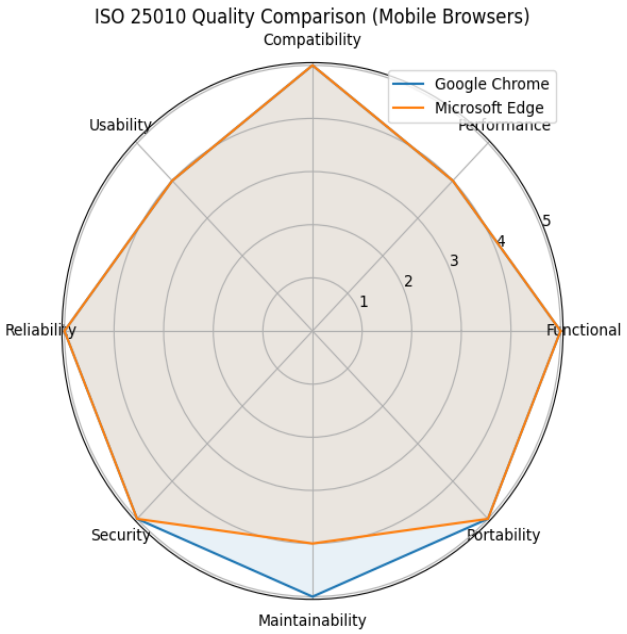
Security: 5 (Strong tracking prevention and privacy controls)

Maintainability: 4 (Regular updates but less frequent than Chrome)

Portability: 5 (Excellent cross-platform operation)

Weighted Score: 4.6 / 5.00

SECTION 5 — Quality Profile Chart



The radar chart visually represents the quality characteristics of both applications, showing nearly identical performance with a slight advantage for Google Chrome in maintainability.

SECTION 6 — Comparative Analysis

- **Google Chrome** achieved a weighted quality score of 4.7/5.00, compared to **Microsoft Edge's** 4.6/5.00, resulting in a small margin of 0.1 points. This indicates that both applications are highly competitive and demonstrate excellent overall quality, with Chrome having a slight advantage in consistency and maintenance.

- The most significant quality gap is observed in Maintainability, where Chrome scores higher due to more frequent updates and faster bug resolution. This provides users with a more stable and continuously improving experience. Another difference appears in Usability, where Edge's additional features introduce slight complexity in navigation, whereas Chrome maintains a simpler interface. Performance Efficiency shows minimal difference, as both browsers deliver fast loading times and acceptable resource usage, making it a neutral factor in comparison.

- A notable trade-off exists between feature richness and usability. Microsoft Edge includes advanced features such as tracking prevention and built-in tools, which enhance functionality and privacy. However, these features can slightly reduce usability due to increased interface complexity. In contrast, Google Chrome prioritizes simplicity and ease of use, which improves usability but limits built-in advanced features.

- Based on this analysis, Google Chrome is recommended for users who prefer a simple, fast, and consistently updated browsing experience. Microsoft Edge is recommended for users who prioritize privacy and additional built-in features. The final choice depends on user requirements, but Chrome holds a slight overall advantage.

SECTION 7 — Recommendations

App 1 Google Chrome

Recommendation 1:
Google Chrome should implement a built-in ad blocker to improve Usability from its current rating of 4 to a target of 5. Currently, users must rely on third-party solutions for blocking ads, which reduces convenience. This change would provide a cleaner browsing experience and improve ease of use.

Recommendation 2:
Google Chrome should optimize battery consumption during active browsing to improve Performance Efficiency from its current rating of 4 to a target of 5. Currently, the application consumes relatively higher battery during extended use. This improvement would enhance efficiency and allow longer usage without frequent charging.

App 2 Microsoft Edge

Recommendation 1:

Microsoft Edge should simplify its user interface by reducing visual clutter and organizing features more clearly to improve Usability from its current rating of 4 to a target of 5. Currently, the presence of multiple features makes navigation slightly complex for users. This change would make the application more intuitive and easier to use.

Recommendation 2:
Microsoft Edge should increase the frequency of software updates and bug fixes to improve Maintainability from its current rating of 4 to a target of 5. Currently, updates are less frequent compared to competitors, which may delay issue resolution. This improvement would ensure faster bug fixes and a more stable user experience.